RESTRICTION GIVES A CANONICAL CONTINUOUS SURJECTION OF GALOIS GROUPS IN A TOWER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The restriction map of this lemma is exactly the map that the exact-sequence restatement (declined unit lemma-ses-infinite-galois) would have packaged.

Editorial note. The label lemma-galois-infinite names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter lemma-lift-maps (delivered card L48358): item 1 restricts automorphisms to normal subextensions; item 2 extends K-maps into a normal extension to automorphisms
- Chapter lemma-galois-profinite (delivered card L48328): part (1) is the universal property of the profinite topology, characterizing continuity of group maps through continuity of the action
- Chapter section setup: Gal(L over K) carries the Krull topology, for which the action Gal(L over K) x L to L is continuous

Source reference. Original reference 0BMJ: sources/stacks-fields/pinned-original-source.tex, source lines 3076–3076. Prerequisite pointer invoked at 3187; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BME: sources/stacks-fields/pinned-original-source.tex, source lines 2163–2163. Prerequisite pointer invoked at 3175; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $L/M/K$ be a tower of fields. Assume both L/K and M/K are Galois. Then there is a canonical surjective continuous homomorphism $c : Gal(L/K) \to Gal(M/K)$.*

Proof. By Lemma 0BME given $\tau : L \to L$ in $\mathrm{Gal}(L/K)$ the restriction $\tau|_M : M \to M$ is an element of $\mathrm{Gal}(M/K)$. This defines the homomorphism c. Continuity follows from the universal property of the topology: the action

$$\mathrm{Gal}(L/K) \times M \longrightarrow M, \quad (\tau, x) \longmapsto \tau(x) = c(\tau)(x)$$

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is continuous as $M \subset L$ and the action $\mathrm{Gal}(L/K) \times L \to L$ is continuous. Hence continuity of c by part (1) of Lemma 0BMJ. Lemma 0BME also shows that the map is surjective. $\square$